



**Fatima Jinnah
Women University**

Department of Computer Science

Data Analytics

**Project Report
BCS-IV(A)**

Instagram Reels Scraping
with and Analysis

Submitted by: Malaika Arooj

Submitted to: Dr. Sadaf

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Table of Contents

Part 1 – Instagram Reels Scraping with Apify

I. Get the Apify API Token

II. Find the Scraper Tool

III. Auto-Scroll Method — Latest Reels

IV. Hardcoded Method — Specific Reels

V. Data Saved in Apify Storage

Part 1 — Conclusion

Part 2 — AI vs Human Sentiment Analysis

I. Methodology

II. AI Tools Used

III. Prompt Used

IV. AI Tool Performance Comparison

V. Which AI Matched the Human Best?

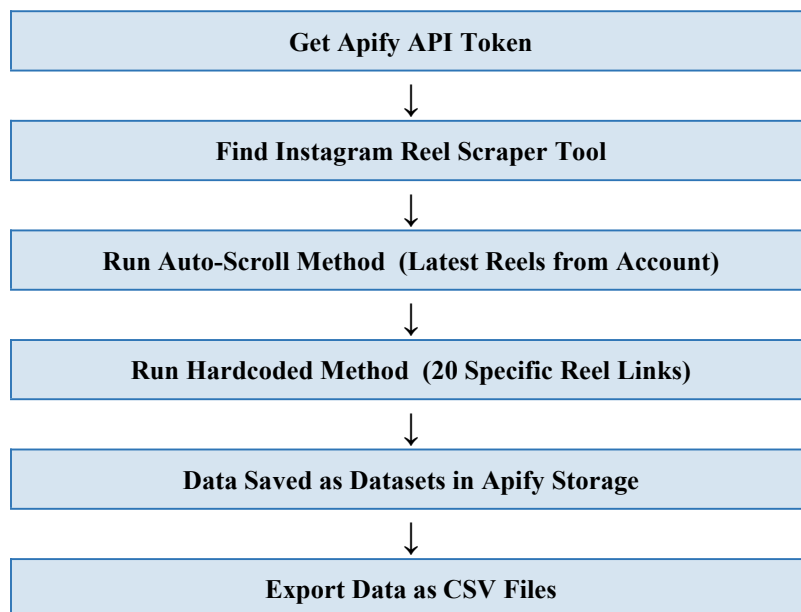
Part 2 — Conclusion

Instagram Reels Data Project

Part 1 — Instagram Reels Scraping with Apify

This part explains how Instagram reels data was collected from an entrepreneur's account using Apify, a web scraping tool. Two methods were used to gather the data: an auto-scroll method that picked up the latest reels automatically, and a hardcoded method that targeted a specific list of reel links.

Project Flow



I. Get the Apify API Token

An account was created on Apify. Then, from the Settings page, the personal API token was copied. This token is used to connect to Apify from the code and run the scraper.

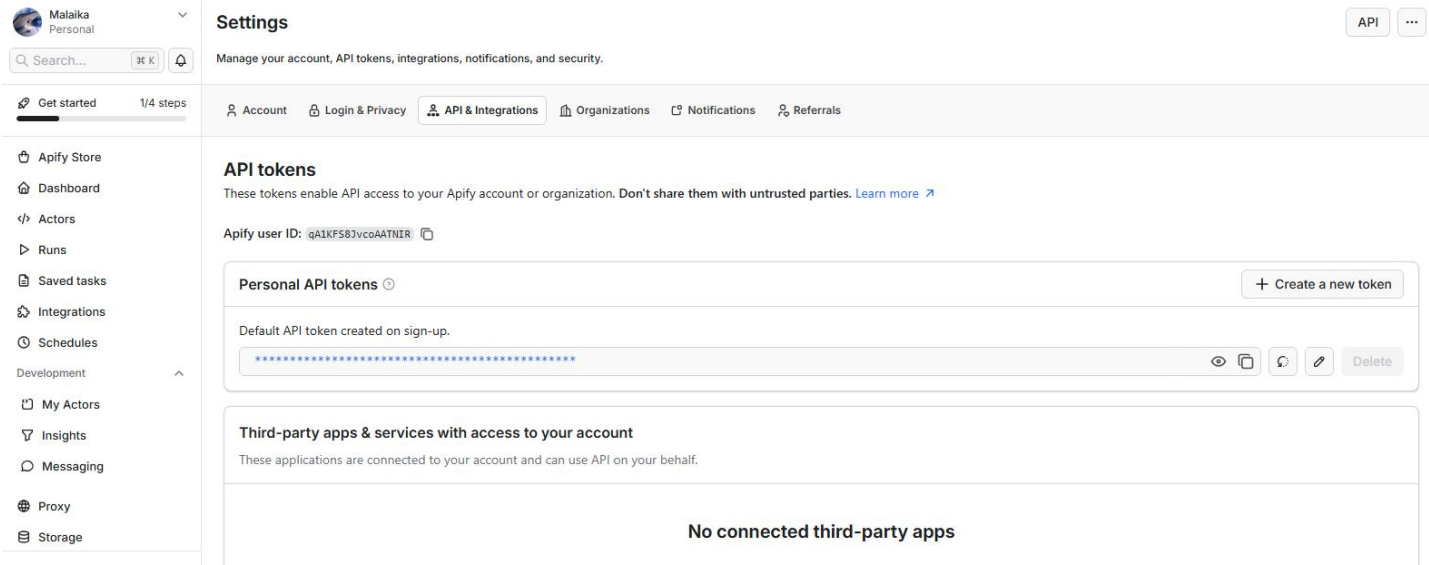


Figure 1: Apify API token in Settings

II. Find the Scraper Tool

The Apify Store was searched for “Instagram Reel Scraper” and the matching tool was selected. This is the actor used to scrape reels data.

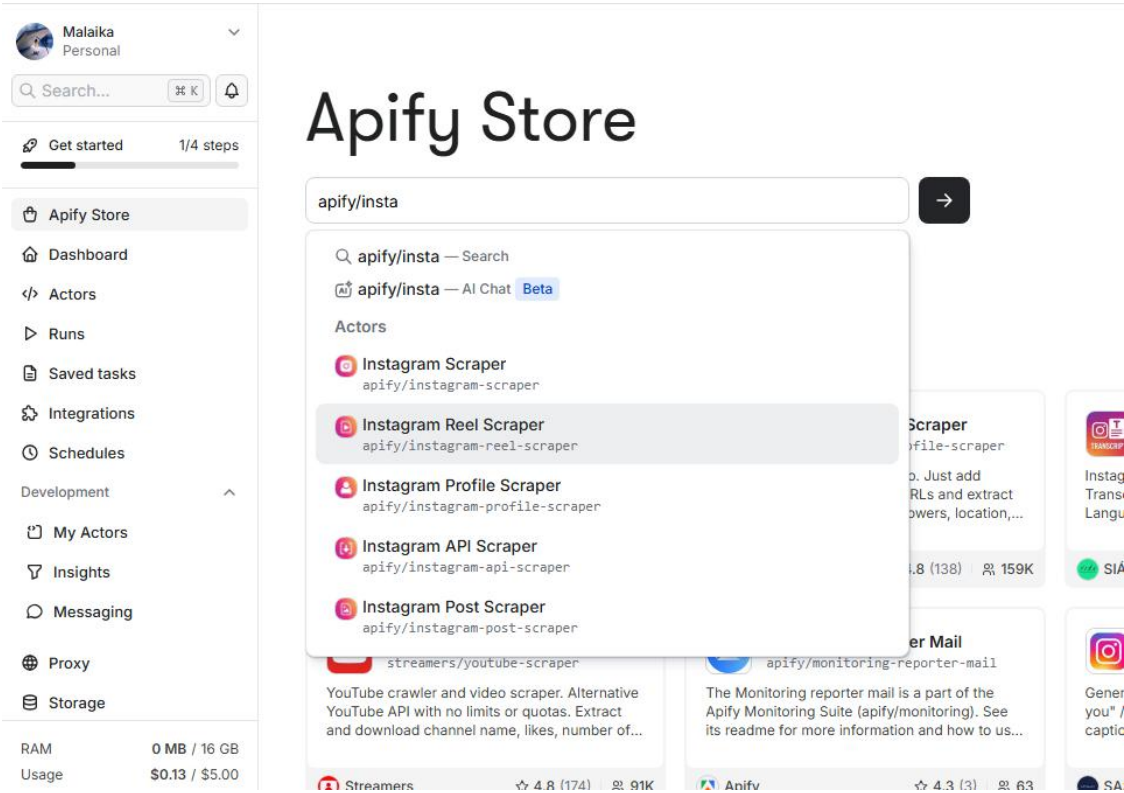
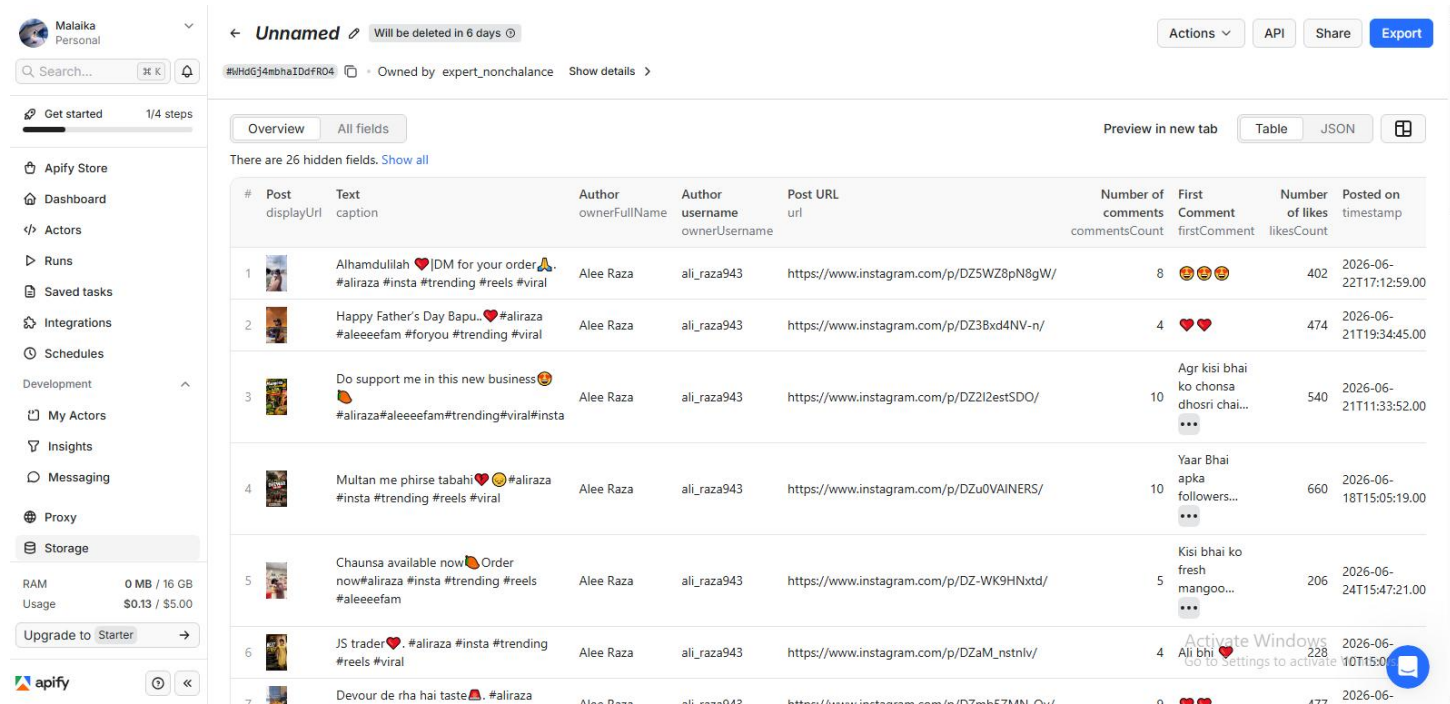


Figure 2: Searching for Instagram Reel Scraper in Apify Store

III. Auto-Scroll Method — Latest Reels

For this method, only one Instagram account link was given to the tool. It automatically scrolled through the account and collected the latest reels on its own. This run collected 30 reels.

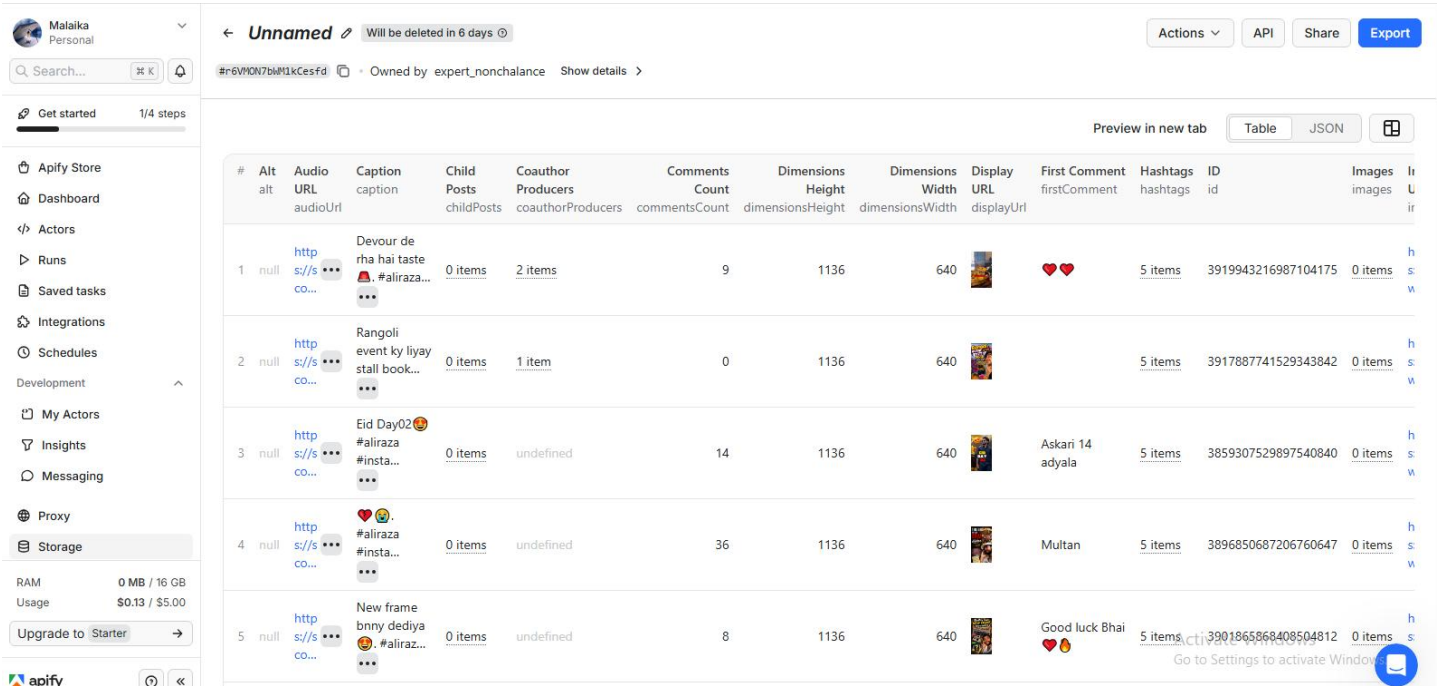


#	Post displayUrl	Text caption	Author ownerFullName	Author username ownerUsername	Post URL url	Number of comments commentsCount	First Comment firstComment	Number of likes likesCount	Posted on timestamp
1		Alhamdulillah ❤️ IDM for your order 🙏. #aliraza #insta #trending #reels #viral	Alee Raza	ali_raza943	https://www.instagram.com/p/DZ5WZ8pN8gW/	8	👍👍👍	402	2026-06-22T17:12:59.00
2		Happy Father's Day Bapu..❤️ #aliraza #aleeefam #foryou #trending #viral	Alee Raza	ali_raza943	https://www.instagram.com/p/DZ3Bxd4NV-n/	4	❤️❤	474	2026-06-21T19:34:45.00
3		Do support me in this new business 🙏 #aliraza #aleeefam #trending #viral #insta	Alee Raza	ali_raza943	https://www.instagram.com/p/DZ2l2estSDO/	10	Agr kisi bhai ko chonsa dhosri chai...	540	2026-06-21T19:34:45.00
4		Multan me phirse tabahi 🙏❤️ #aliraza #insta #trending #reels #viral	Alee Raza	ali_raza943	https://www.instagram.com/p/DZu0VAINERS/	10	Yaar Bhai apka followers...	660	2026-06-18T15:05:19.00
5		Chaunsa available now 🙏 Order now #aliraza #insta #trending #reels #aleeefam	Alee Raza	ali_raza943	https://www.instagram.com/p/DZ-WK9HNxtD/	5	Kisi bhai ko fresh mango...	206	2026-06-24T15:47:21.00
6		JS trader ❤️ #aliraza #insta #trending #reels #viral	Alee Raza	ali_raza943	https://www.instagram.com/p/DZaM_nstnlv/	4	Ali bhi ❤️	228	2026-06-10T15:05:19.00
7		Devour de rha hai taste 🙏. #aliraza	Alee Raza	ali_raza943	https://www.instagram.com/p/DZ7mh57MNL_Cu/	9	❤️❤	177	2026-06-

Figure 3: Latest reels fetched automatically from one account

IV. Hardcoded Method — Specific Reels

For this method, 20 specific reel links were chosen manually and given to the tool. It fetched data only for those 20 reels.

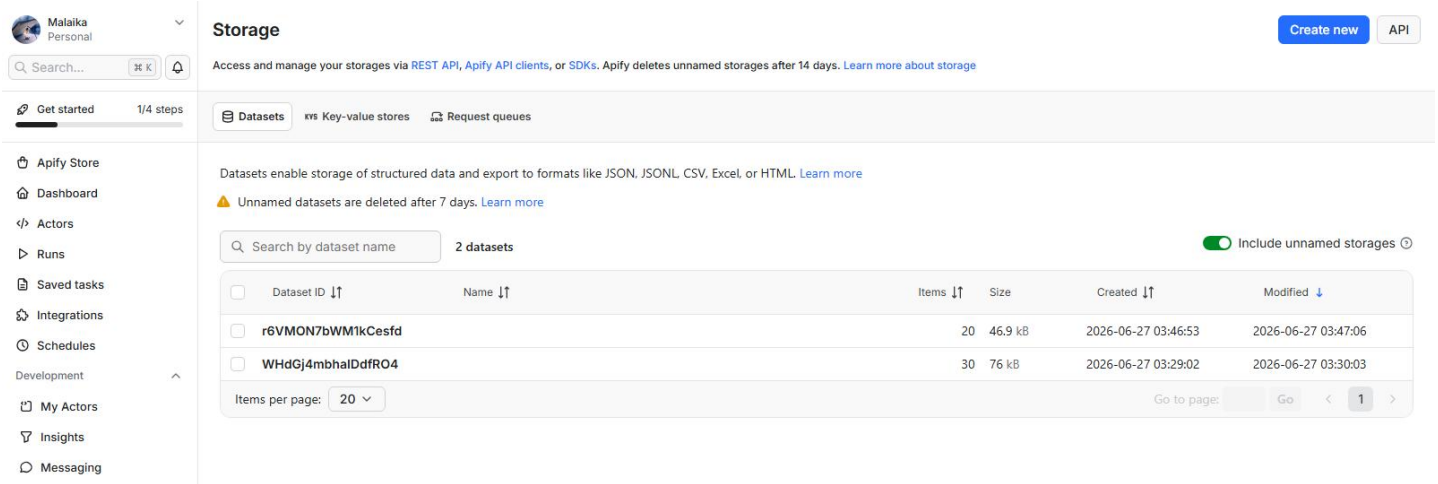


#	Alt	Audio URL	Caption	Child Posts	Coauthor Producers	Comments Count	Dimensions Height	Dimensions Width	Display URL	First Comment	Hashtags	ID	Images
1	null	http://s/s/co...	Devour de rha hai taste . #aliraza...	0 items	2 items	9	1136	640		❤️ ❤️	5 items	3919943216987104175	0 items
2	null	http://s/s/co...	Rangoli event ky liyay stall book...	0 items	1 item	0	1136	640			5 items	3917887741529343842	0 items
3	null	http://s/s/co...	Eid Day02 #aliraza #insta...	0 items	undefined	14	1136	640		Askari 14 adyala	5 items	3859307529897540840	0 items
4	null	http://s/s/co...	❤️🥳. #aliraza #insta...	0 items	undefined	36	1136	640		Multan	5 items	3896850687206760647	0 items
5	null	http://s/s/co...	New frame bnny dediya 🥳. #aliraza...	0 items	undefined	8	1136	640		Good luck Bhai ❤️🔥	5 items	3901865868408504812	0 items

Figure 4: Data fetched for 20 specific reel links

V. Data Saved in Apify Storage

Both runs saved their results as datasets inside Apify Storage. The Dataset ID of each run was used to pull the data and save it as CSV files.



Dataset ID	Name	Items	Size	Created	Modified
r6VMON7bWM1kCesfd		20	46.9 kB	2026-06-27 03:46:53	2026-06-27 03:47:06
WHdGj4mbhaIdDfR04		30	76 kB	2026-06-27 03:29:02	2026-06-27 03:30:03

Figure 5: Two datasets saved in Apify Storage

Part 1 — Conclusion

Instagram reels data was collected using two methods: auto-scroll (latest reels from an account) and hardcoded links (20 specific reels chosen manually). Both results were saved as CSV files for further analysis.

Part 2 — AI vs Human Sentiment Analysis

This part compares how a human reads the sentiment of social media comments versus how five AI tools (LLMs) read the same comments. The dataset used here is 50 comments collected from one of the reels gathered in Part 1, written mostly in Roman Urdu mixed with English.

i. Methodology

All 50 comments were read manually and given one of three labels: Positive, Negative, or Neutral. The same comments were then sent to five AI tools using one identical prompt so the comparison would be fair.

ii. AI Tools Used

- ChatGPT — GPT-4o (OpenAI)
- Gemini — Gemini 1.5 Pro (Google)
- Claude — Claude 3.5 Sonnet (Anthropic)
- Meta AI — Llama 3 (Meta)
- Grok — Grok-2 (xAI)

iii. Prompt Used

“I am analyzing a social media comment written in Roman Urdu and/or English. Classify the sentiment of the comment as Positive, Negative, or Neutral. Treat prayers, blessings, praise, encouragement, and other kind or supportive language as Positive. Treat criticism, complaints, boycott calls, or ethical objections — even if the wording sounds harsh — as Negative. Treat plain questions, suggestions, or observations that carry no clear emotional charge as Neutral. Comment: [comment text]”

II. AI Tool Performance Comparison

The table below ranks each AI tool by how many of the 50 comments it labelled the same way as the human reviewer.

AI Tool	Version	Correct	Accuracy	Rank
Grok (xAI)	Grok-2	48/50	96%	1st
Claude (Anthropic)	Claude 3.5 Sonnet	47/50	94%	2nd
Gemini (Google)	Gemini 1.5 Pro	44/50	88%	3rd
Meta AI (Meta)	Llama 3	38/50	76%	4th
ChatGPT (OpenAI)	GPT-4o	37/50	74%	5th

III. Which AI Matched the Human Best?

Grok-2 agreed with the human reviewer on 48 out of 50 comments (96%), making it the closest match. Claude 3.5 Sonnet came second at 94%, followed by Gemini 1.5 Pro at 88%. ChatGPT and Meta AI struggled the most, mainly because they labelled boycott calls and ethical objections as Positive instead of recognizing the criticism underneath.

All five AI tools had the most trouble with comments that mixed support and criticism together, such as “We support you but don’t support genocide.” A human reader reads the second part as the real meaning of the comment, while most AI tools focus only on the friendly opening words.

Part 2 — Conclusion

Grok-2 and Claude 3.5 Sonnet were the strongest matches to human judgement for this bilingual, culturally specific dataset. For Pakistani social media comments written in Roman Urdu, the most practical approach is to let Grok or Claude do the first pass, then have a human spot-check the Neutral and Negative labels, especially comments that mention a brand, a boycott, or an ethical concern.